

To see why cool, saturated air near the surface allows for only elevated convection, use the slider on the right to see how parcels at various levels might be lifted (due to overrunning for example). Since the parcels are already saturated, they follow a moist adiabat. Buoyancy is denoted by the color of the parcel's path (blue - negative; orange - positive). Notice that only parcels near the top of the inversion are positively buoyant.

